

Dense matrix block-encoding: Clader-Frobenius vs SVD interferometer

Task: block-encode a single dense $N \times N$ matrix of classical data, $N = 2^n$,
 $n = 4..10$, `phase_bitsize = 32`. Two methods, each
in a Toffoli(T)-count-optimal and a qubit-count-optimal configuration.

(1) CLADER-FROBENIUS (arXiv:2206.03505, minimal-T-count construction)

$U_A = U_R^\dagger U_L$, subnormalization $\alpha = ||A||_F$.

U_L = plain state prep of $|\phi\rangle = \sum_j (|A_j|/||A||_F) |j\rangle$ (+ register swap)

U_R = controlled state prep of $|\psi_j\rangle = \sum_k (A_{jk}/|A_j|) |k\rangle$

Both are select-swap / QROAM-rotation state preparations (Low-Kliuchnikov-Schaeffer phase-gradient rotations).

T-opt : per-layer analytic λ * QROAM batching (fewest T, more ancilla).

Q-opt : un-batched $\lambda = 1$ QROAM (fewest qubits, more T).

(2) SVD INTERFEROMETER (subnormalization $\alpha = 1$)

$A = U \Sigma V$; U, V via the block-unitary interferometer ($n_{\text{blocks}} = 1$), the

diagonal Σ via QROAMClean + ctrl-AddIntoPhaseGrad + QROAMCleanAdjoint.

Each metric is optimized over a uniform QROAM batching λ in $\{0..n\}$.

SCALING FITS $y = c * 2^{(a*n)}$ ($a \sim$ slope per doubling of n):

Clader-Frobenius, Toffoli-opt	T: $a=1.00$ $c=5.23e+01$	Q: $a=0.88$ $c=8.72e+00$
Clader-Frobenius, qubit-opt	T: $a=1.93$ $c=4.55e+00$	Q: $a=0.06$ $c=6.85e+01$
SVD interferometer, Toffoli-opt	T: $a=1.29$ $c=1.04e+02$	Q: $a=0.25$ $c=4.31e+01$
SVD interferometer, qubit-opt	T: $a=1.37$ $c=7.82e+01$	Q: $a=0.03$ $c=9.58e+01$

KEY CAVEAT - subnormalization is NOT equal:

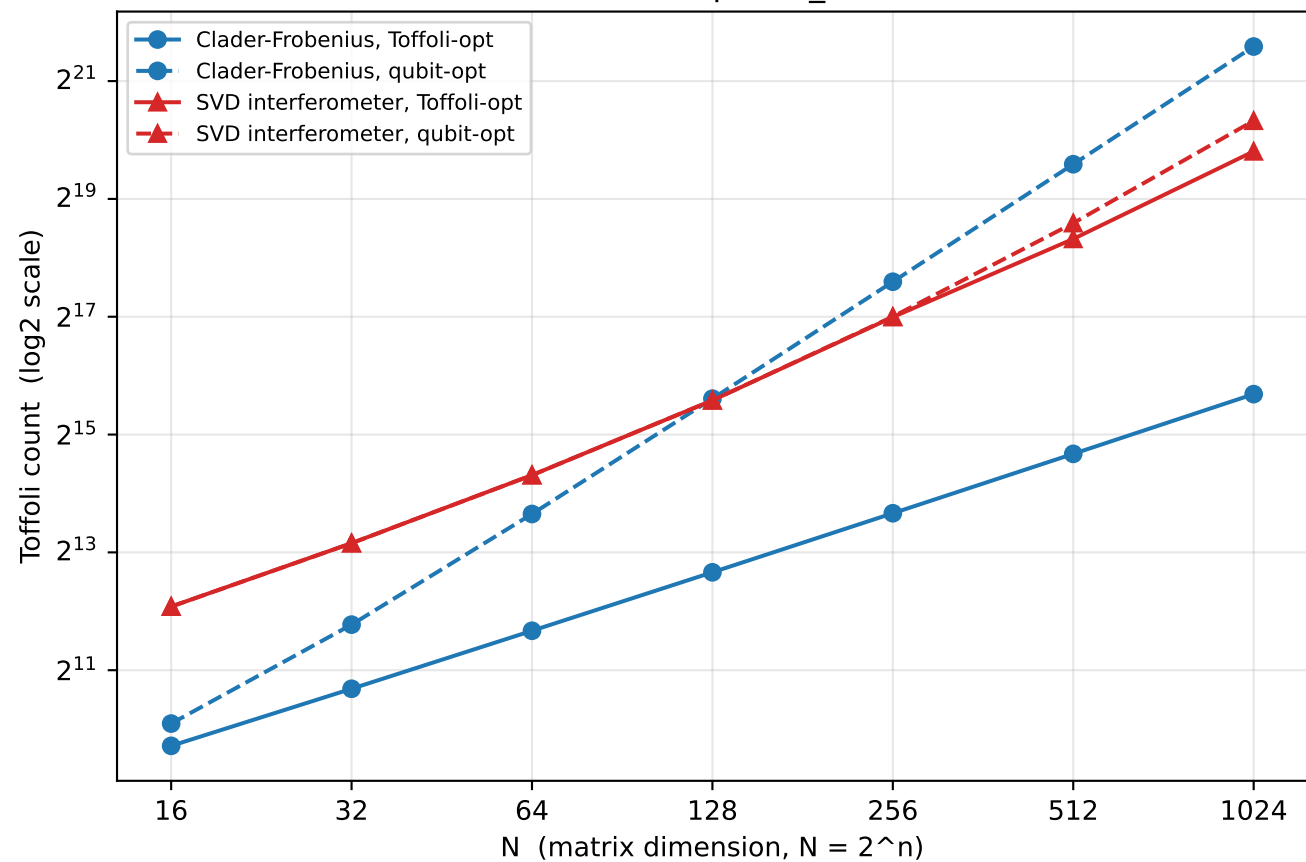
- * Clader-Frobenius gives $\alpha = ||A||_F$, which is $\sim N$ for a dense matrix with $O(1)$ entries. SVD interferometer gives the spectral-norm-optimal $\alpha = 1$.
- * Downstream algorithms (QSVT, QPE, qubitization) require $\sim \alpha$ block-encoding queries. So the EFFECTIVE Clader cost for such an algorithm is roughly its per-query Toffoli count times $||A||_F$ ($\sim N$), whereas the SVD cost is per-query.
- * Read the raw T/Q here as the cost of ONE block-encoding query. The SVD method pays more per query but needs far fewer queries; Clader is cheap per query but its large α must be amortized. The right choice is application-dependent.

Both constructions: error $\sim O(2^{-\text{phase_bitsize}})$ from angle discretization; counts
from Qualtran `QECGatesCost / QubitCount` (`utils.get_Toffoli_counts / get_qubit_counts`).

Toffoli (T) and qubit (Q) counts: dense N x N block-encoding
 phase_bitsize = 32; alpha(Clader) = ||A||_F, alpha(SVD) = 1

n	N	Clader-Frobenius Toffoli-opt	Clader-Frobenius qubit-opt	SVD interferometer Toffoli-opt	SVD interferometer qubit-opt
4	16	T=841 Q=112	T=1,093 Q=81	T=4,329 Q=103	T=4,329 Q=103
5	32	T=1,646 Q=179	T=3,501 Q=85	T=9,129 Q=105	T=9,129 Q=105
6	64	T=3,257 Q=310	T=12,869 Q=89	T=20,329 Q=107	T=20,329 Q=107
7	128	T=6,480 Q=569	T=49,981 Q=93	T=49,001 Q=109	T=49,001 Q=109
8	256	T=12,983 Q=1,084	T=197,877 Q=97	T=130,508 Q=174	T=131,177 Q=111
9	512	T=26,110 Q=2,111	T=788,525 Q=101	T=327,372 Q=176	T=394,345 Q=113
10	1024	T=52,725 Q=4,162	T=3,149,413 Q=105	T=917,394 Q=305	T=1,314,921 Q=115

Toffoli count vs N (phase_bitsize = 32)



Qubit count vs N (phase_bitsize = 32)

